MILITARY SPECIFICATION
FIRING MECHANISM, 50057-1: OVERHAUL OF

1. SCOPE:

1.1 This specification covers the overhaul requirements for Firing Mechanism, 50057-1. The assembly will be used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS:

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	Ammunition and Special Weapons, General Specification for
MIL-L-4343	Lubricating Grease, Pneumatic System

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-1167	Ammunition Data Card
MS-28741	Hose Assembly- Detachable End Fitting, Medium Pressure.

DRAWINGS

Munitions Command

C9221280	Container, Ammunition, Fiber for Firing Mechanism Assy, P/N 50057-1.
C9221282	Box, Packing, Ammunition, for Firing Mechanism Assy, P/N 50057-1.

PUBLICATIONS

Air Force Technical Manual

T. O. 11P8-4-3	Overhaul Instructions with Parts Breakdown, Firing Mechanism Assy, 50057-0 and 50057-1.

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(Copies of specifications, standards and technical manuals required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with requirements specified in Technical Order 11P8-4-3 and referenced specifications.

3.2 Assembly torque.- The Chamber, P/N 50009-0 shall be assembled to the Head, P/N 50005 with an assembly torque as specified in Technical Order 11P8-4-3.

3.3 The overhauled Firing Mechanism Assembly shall contain new parts in accordance with Table I.

TABLE I

Name of Part	Part Number
Pin, Shear	50007-0
O-Ring	AN 6227-7
Gasket	AN6290-4
Set Screw	AN565D8H2

3.4 Protective finish.- Protective finish shall be in accordance with Technical Order 11P8-4-3.

3.5 Residual Magnetism.- The firing mechanism shall not deflect the indicator of the approved compass more than 3 degrees in either direction when passed in front of the approved compass at a distance of 6 inches.

3.6 Functional

3.6.1 The firing mechanism shall not fail to fire

3.6.2 The firing mechanism shall function without mechanical failure (6.1).

3.6.3 Initiation.- The firing mechanism shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size 4, standard MS-28741, at 70° ± 5°F.

3.6.4 Ballistics.- The firing mechanism loaded with an accepted cartridge and temperature conditioned at 70°F ± 5°F shall be fired into a 5.513 cubic inch volume for pressure-time performance and data recorded for information.

3.7 Workmanship

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3.7.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defects.

3.7.2 Burr.- Parts shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- Parts or the assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies provided the homogeneity of the rebuild process has been retained. The assembly shall be loaded with a single lot of explosive components.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. Functional testing shall be utilized in lieu of X-ray examination.

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4.2.2.1 Reclaimed parts.- Examination of reusable parts, which require no reprocessing shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be rejected.

4.2.2.2 Reprocessed parts.- One hundred percent examination shall be performed for all characteristics. Examination shall be limited to visual type inspection unless otherwise specified. All nonconforming material shall be rejected.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over - 43	6	8	11	14
Sample Sizes for Minor Defects	**110**	**150**	**225**	**300**
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over - 48	8	11	17	20

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4.2.2.3 Cartridge Assembly, P/N 50005, covering a detail of P/N 50057-1

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Sealing disc missing or damaged	Visual
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Protective finish, missing or inadequate	Visual

4.2.2.4 Assembly, P/N 50057-1, less P/N AN565D8H2

CRITICAL	
1. Firing pin missing	Visual
2. Shear pin missing or not in groove	Visual
MAJOR	
None defined	
MINOR	
None defined	

4.2.2.5 Assembly P/N 50057-1

CRITICAL	
1. Improper assembly	Visual
2. Any parts missing	Visual
3. Labels missing, incorrect or illegible	Visual
MAJOR	
101. Proper alignment	Visual
102. Set screw missing, loose, above flush, or not covered with sealing compound	Visual

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4.2.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Location of labels incorrect	Visual
202. Protective finish missing or inadequate	Visual
203. Shipping cap missing	Visual

4.2.2.6 Container, Ammunition, Fiber; Drawing C9221280

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking, missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.7 Box, Packing, Ammunition, Drawing C9221282

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken, or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual

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4.2.2.7 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
204. Contents of box exposed	Visual
205. Marking missing, incorrect or illegible	Visual
206. Number of units incorrect	Visual
207. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Assembly torque.- Threaded joints of each assembly shall be subjected to the torque application. Failure of any assembly to comply with the requirement of 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Protective finish, replated components.- The number of specimens and acceptance criteria for protective finish shall be in accordance with the applicable Military Standard for the type of protective finish utilized in the rebuild process.

4.2.3.3 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.5 shall be cause for rejection of the assembly.

4.2.3.4 Functional tests.- The following tests shall be performed by the Government.

4.2.3.4.1 Ballistics.- Twenty-one units shall be tested as specified in 4.3.4 to determine compliance with the requirements of 3.6.1 to 3.6.4. The units shall be temperature conditioned at 65° to 75°F as provided in 4.3.4.1.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment approved by the Government.

4.3 Test methods and procedures

4.3.1 Assembly torque.- A torque wrench, set at the torque requirements specified in 3.2 shall be used in conjunction with an adapter to determine the adequacy of the joints. The firing mechanism shall be secured in a holding device and the assembly torque applied to the head.

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4.3.2 Protective finish, replated components.-Test procedures for protective finish shall be in accordance with the applicable Military Standards for the type of protective finish utilized in the rebuild process.

4.3.3 Residual magnetism.-The assembly shall be tested for excessive residual magnetism, in an area free of local magnetic effects, by placing the assembly six inches from and at the same height as the compass in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly past the compass. This procedure shall be repeated for each 90 degree rotation of the firing mechanism.

4.3.4 Functional tests.

4.3.4.1 Temperature conditioning.-Prior to conducting the ballistic test specified in 4.3.4.2, the assembly samples shall be subjected to the temperature condition specified in 3.6.4. A simulated assembly loaded with a dummy cartridge containing a temperature sensitive element shall be placed in the temperature conditioning chamber at the same time as the assembly samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.3.4.1.1 Upon completion of the required temperature conditioning, each assembly sample shall be ballistically tested within 5 minutes after removal from the conditioning chamber. Samples shall be reconditioned if this time is exceeded.

4.3.4.2 Method of testing.-The firing mechanism shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size No. 4, MS 28741 to determine compliance with 3.6. The firing mechanism shall be rigidly supported in a test fixture and fired into a 5.513 cubic inch volume for pressure time performance.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.-Packing and marking shall be in accordance with drawings C9221280 and C9221282.

5.2 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

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